2. Solution:

~~Sta~~ equation of clock: $CP_0 = CP_0$, $CP_1 = \bar{Q}_0$ $CP_2 = \bar{Q}_1$ $CP_3 = \bar{Q}_0$

equation of ~~state~~ driver: FF₀: $\cancel{Q_0^{n+1} = \overline{Q_0^n}}$, $\cancel{CP_0}\downarrow$ $J_0 = K_0 = 1$ $CP_0 \downarrow$

FF₁: $J_1 = \overline{\bar{Q}_2 \bar{Q}_3}$ $K_1 = 1$ $\bar{Q}_0 \downarrow$. $Q_0 \uparrow$

FF₂: $J_2 = K_2 = 1$ $\bar{Q}_1 \downarrow$. $Q_1 \uparrow$

FF₃: $J_3 = \bar{Q}_1 \bar{Q}_2$ $K_3 = 1$. $\bar{Q}_0 \downarrow$. $Q_0 \uparrow$

equation of state: FF₀: $Q_0^{n+1} = \bar{Q}_0^n$ $CP_0 \downarrow$

FF₁: $Q_1^{n+1} = (Q_2 + Q_3)\bar{Q}_1^n$ $Q_0 \uparrow$

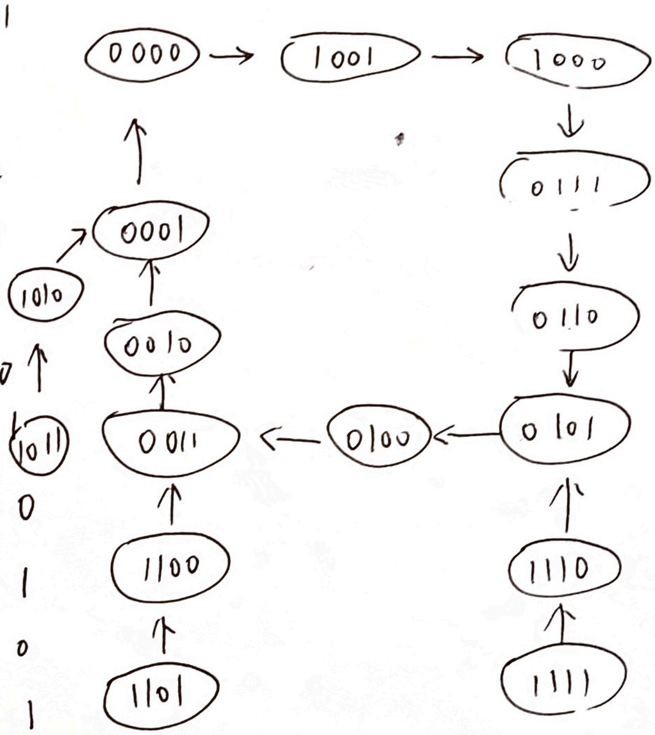
FF₂: $Q_2^{n+1} = \overline{Q_2^n}$ $Q_1 \uparrow$

FF₃: $Q_3^{n+1} = \bar{Q}_1 \bar{Q}_2 \bar{Q}_3$ $Q_0 \uparrow$

equation of output: $CO = \overline{\bar{Q}_3 \bar{Q}_2 \bar{Q}_1 \bar{Q}_0} = Q_0 + Q_1 + Q_2 + Q_3$

Transformation table

Q_3^n	Q_2^n	Q_1^n	Q_0^n	Q_3^{n+1}	Q_2^{n+1}	Q_1^{n+1}	Q_0^{n+1}
0	0	0	0	1	0	0	1
~~0~~	~~0~~	~~0~~	~~1~~				~~0~~
1	0	0	1	1	0	0	0
1	0	0	0	0	1	1	1
0	1	1	1	0	1	1	0
0	1	1	0	0	1	0	
0	1	0	1	0	1	0	0
0	1	0	0	0	0	1	1
0	0	1	1	0	0	1	0
0	0	1	0	0	0	0	1
0	0	0	1	0	0	0	0
1	0	1	0	0	0	0	1
1	0	1	1	1	0	1	0
1	1	0	0	0	0	1	1
1	1	0	1	1	1	0	0
1	1	1	0	0	1	0	1
1	1	1	1	1	1	1	0



Function:

Asychronous, Decimal substract counter. Self-starting

3. Solution:

Clock: $CP = CP$

~~preset~~ $D_3 D_2 D_1 D_0 = 0000$

$\overline{CR} = \overline{Q_1 Q_3}$, $CO = Q_3$

$M = 1010 = 10.$

Use the Asychronous - clear method , Decimal counter.

4. solution:

The 74161(I), 74161(II) construct a 16×16 - ary counter.

clock: $CP = CP$

preset: 0

$\overline{LD} = \overline{Q_{2II} Q_{0II} Q_{1I}}$, $CO = \overline{LD}$

$\cancel{M = 2 \times 5 = 10}$ $\cancel{M = 11}$ $M-1 = 5 \times 16 + 2 \times 16^0$ $M = 83$

use the sychronous ~~method~~ preset methol. 83- ary counter.

Ch07 Assignments Digital Logic

1120230272 兰天逐

1. Analysis the function of the sequential logic circuit.

Solution: equation of clocks: $CP_1 = CP$, $CP_2 = CP$. $CP_3 = CP$. sychronous.

equation of driver: FF_1: $J_1 = \bar{Q}_3$ $K_1 = \bar{Q}_3$

FF_2: $J_2 = Q_1$ $K_2 = Q_1$

FF_3: $J_3 = Q_1 Q_2$ $K_3 = Q_3$

equation of output: $CO = Q_3$

equation of state: $Q^{n+1} = J\bar{Q}^n + \bar{K} Q^n$.

FF_1: $Q_1^{n+1} = \bar{Q}_3 \bar{Q}_1^n + Q_3^n Q_1^n = Q_3^n \odot Q_1^n$

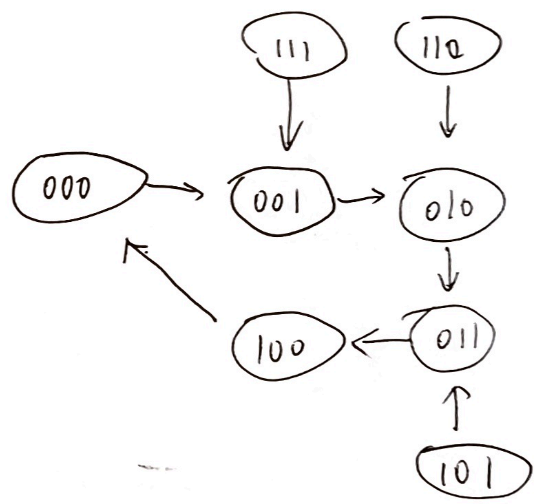
$Q_2^{n+1} = Q_1^n \bar{Q}_2^n + \bar{Q}_1 Q_2^n = Q_2^n \oplus Q_1^n$

$Q_3^{n+1} = Q_1^n Q_2^n \bar{Q}_3^n + \bar{Q}_3 Q_3^n = \cancel{(\bar{Q}_1^n + \bar{Q}_2^n) \bar{Q}_3^n}$

$Q_1^n Q_2^n \bar{Q}_3^n$

State transformation table.,

Q_3^n	Q_2^n	Q_1^n	Q_3^{n+1}	Q_2^{n+1}	Q_1^{n+1}
0	0	0	0	0	1
0	0	1	0	1	0
0	1	0	0	1	1
0	1	1	1	0	0
1	0	0	0	0	0
1	0	1	0	1	1
0	1	1	1	0	0
1	0	0	0	0	0
1	1	0	0	1	0
1	1	1	0	0	1



Function:

Sychronous, 5-ary additional counter

self-starting